Supplementary Materials: Critical Slowing Down Indicators for LLM Capability Boundary Detection

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1 Appendix A: Per-Model-Task CSD Indicator Profiles

This appendix presents the full CSD indicator profiles for each model-task pair that exhibits a sharp capability boundary (d^*). For each pair, we report all CSD indicators (embedding variance, Hartigan dip, silhouette k=2, bimodality coefficient, disagreement rate) across all difficulty levels.

1.1 arithmetic – llama-3.1-8b-instruct ($d^* = 20$)

d	Acc	Variance	Dip	Silhouette	BC	Disagree
2	0.6000	0.3109	0.1243	0.3512	0.6224	0.8600
3	0.6200	0.2256	0.1503	0.2998	0.7329	0.8200
4	0.7800	0.2494	0.1032	0.2513	0.6470	0.8000
5	0.6400	0.1967	0.0704	0.2432	0.6249	0.8400
6	0.7800	0.2538	0.0631	0.2948	0.3968	0.8000
7	0.7400	0.2175	0.0730	0.3125	0.6417	0.8200
8	0.7600	0.2417	0.1104	0.2501	0.6613	0.8200
9	0.7800	0.2074	0.0740	0.2335	0.5718	0.8200
10	0.6800	0.1718	0.0699	0.2906	0.6429	0.8400
11	0.6000	0.1604	0.0763	0.2505	0.5909	0.8600
12	0.7200	0.2011	0.0905	0.2489	0.6034	0.8200
13	0.5600	0.1645	0.0614	0.2009	0.5526	0.8600
14	0.7143	0.1851	0.0651	0.2580	0.6192	0.8095
15	0.7800	0.1627	0.0676	0.2751	0.5327	0.8200
16	0.7000	0.1741	0.0690	0.2895	0.5594	0.8400
17	0.7000	0.1411	0.0853	0.3600	0.6602	0.8200
18	0.6000	0.1922	0.0515	0.2931	0.7161	0.8400
19	0.7000	0.2381	0.0787	0.2891	0.7292	0.8000
20 †	0.4600	0.1892	0.0958	0.2199	0.5384	0.8400
21	0.6400	0.1839	0.1069	0.2340	0.4572	0.8200
22	0.5800	0.2007	0.0545	0.2313	0.6315	0.8000
23	0.7400	0.1881	0.0867	0.2588	0.6670	0.8400
24	0.8200	0.1826	0.0722	0.2161	0.5336	0.8200
25	0.8000	0.1932	0.0544	0.2629	0.5458	0.8000

Table 1: CSD indicators for llama-3.1-8b-instruct on arithmetic. † marks $d^* = 20$.

1.2 arithmetic – gemini-2.0-flash-001 ($d^* = 15$)

1.3 arithmetic – gpt-4o-mini ($d^* = 2$)

1.4 graph coloring – gpt-4o-mini ($d^* = 10$)

1.5 graph coloring – gemini-2.0-flash-001 ($d^* = 14$)

1.6 graph coloring – gemini-2.0-flash-lite-001 ($d^* = 11$)

d	Acc	Variance	Dip	Silhouette	BC	Disagree
2	0.5000	0.3051	0.1551	0.4149	0.7056	0.8600
3	0.5600	0.2206	0.1150	0.3119	0.7454	0.8400
4	0.6200	0.2263	0.1310	0.3252	0.6426	0.8400
5	0.5200	0.1868	0.0778	0.2543	0.6919	0.8400
6	0.5600	0.2486	0.0875	0.3574	0.5000	0.8600
7	0.5800	0.2036	0.0858	0.3806	0.7358	0.8000
8	0.6600	0.2440	0.1058	0.2703	0.5117	0.8400
9	0.6200	0.2119	0.0844	0.2756	0.5956	0.8600
10	0.5400	0.1755	0.0893	0.2697	0.6971	0.8400
11	0.5400	0.1755	0.0680	0.2938	0.8548	0.8600
12	0.5200	0.2186	0.1130	0.2387	0.6381	0.8800
13	0.7000	0.1396	0.0577	0.2026	0.4734	0.8200
14	0.6400	0.1883	0.0909	0.2747	0.6548	0.8200
15 †	0.4500	0.1666	0.0557	0.3045	0.7413	0.8250
16	0.6200	0.1469	0.0484	0.1817	0.4858	0.8400
17	0.7200	0.1655	0.0758	0.2482	0.4812	0.8200
18	0.7200	0.1711	0.0723	0.2481	0.5588	0.8400
19	0.5200	0.2318	0.0719	0.3162	0.7805	0.8400
20	0.5400	0.1811	0.0764	0.2067	0.4961	0.8600
21	0.5200	0.1734	0.0783	0.2549	0.5684	0.8800
22	0.7600	0.1611	0.0702	0.3175	0.5254	0.8000
23	0.5000	0.1696	0.0710	0.2459	0.6085	0.8200
24	0.6400	0.1605	0.0710	0.2173	0.5300	0.8400
25	0.6800	0.1698	0.0648	0.2480	0.6191	0.8000

Table 2: CSD indicators for gemini-2.0-flash-001 on arithmetic. † marks $d^* = 15$.

d	Acc	Variance	Dip	Silhouette	BC	Disagree
2 †	0.0000	0.2321	0.1488	0.3475	0.6984	0.8000
3	0.1200	0.2262	0.0833	0.2510	0.6722	0.8000
4	0.0200	0.2260	0.1240	0.2337	0.6724	0.8000
5	0.0800	0.2166	0.0533	0.1740	0.4156	0.8000
6	0.0800	0.2666	0.0700	0.3110	0.6384	0.8000
7	0.2400	0.2601	0.0451	0.2864	0.6000	0.8400
8	0.3000	0.2758	0.0582	0.2047	0.5161	0.7800
9	0.2200	0.2354	0.0337	0.2481	0.5871	0.8200
10	0.3800	0.1967	0.0349	0.1959	0.3464	0.8600
11	0.2800	0.1938	0.0495	0.2523	0.5226	0.8800
12	0.3600	0.1819	0.1059	0.2830	0.7585	0.8600
13	0.2600	0.1401	0.0432	0.1239	0.4181	0.8400
14	0.2000	0.1893	0.0895	0.2518	0.6338	0.8400
15	0.2600	0.1691	0.0439	0.2085	0.3772	0.8600
16	0.2800	0.1415	0.0621	0.1710	0.4067	0.8600
17	0.3000	0.1705	0.0496	0.2111	0.6000	0.8800
18	0.2800	0.1917	0.1199	0.2408	0.6610	0.8200
19	0.3800	0.2132	0.0644	0.2317	0.6088	0.8600
20	0.4200	0.1696	0.0671	0.2099	0.5448	0.8600
21	0.3000	0.1895	0.0741	0.1828	0.5181	0.9000
22	0.2600	0.1812	0.0470	0.1953	0.4605	0.8800
23	0.3200	0.1577	0.0487	0.1607	0.3357	0.8800
24	0.2800	0.1647	0.0444	0.2251	0.6160	0.8600
25	0.3800	0.1535	0.0645	0.2477	0.8133	0.8800

Table 3: CSD indicators for gpt-4o-mini on arithmetic. † marks $d^* = 2$.

d	Acc	Variance	Dip	Silhouette	BC	Disagree
1	1.0000	0.0940	0.0495	0.1651	0.7115	0.3800
2	1.0000	0.1059	0.0278	0.1772	0.5858	0.6400
3	0.7600	0.1025	0.0446	0.1797	0.7561	0.7000
4	0.8600	0.1074	0.0352	0.1388	0.6103	0.8400
5	0.6000	0.0970	0.0539	0.1726	0.7047	0.8600
6	0.6400	0.1022	0.0428	0.1509	0.6926	0.7800
7	0.6800	0.0929	0.0536	0.1568	0.7264	0.9000
8	0.5200	0.1121	0.0417	0.1570	0.5531	0.8000
9	0.6800	0.1037	0.0358	0.1277	0.2998	0.9000
10 †	0.3200	0.1097	0.0373	0.1804	0.7256	0.8800
11	0.2400	0.1126	0.0314	0.1325	0.6853	0.9000
12	0.4000	0.0999	0.0342	0.1100	0.4359	0.9200
13	0.0600	0.1046	0.0497	0.1362	0.3600	0.8800
14	0.1000	0.1053	0.0267	0.1214	0.3946	0.9200
15	0.4600	0.1092	0.0545	0.1382	0.5864	0.9400
16	0.1400	0.1159	0.0482	0.1282	0.7406	0.9400
17	0.3600	0.1099	0.0371	0.1383	0.6275	0.9600
18	0.2600	0.1180	0.0401	0.1337	0.6021	0.9400
19	0.1600	0.1202	0.0302	0.1481	0.4539	0.9600
20	0.0600	0.1232	0.0443	0.1285	0.3368	0.9600

Table 4: CSD indicators for gpt-4o-mini on graph coloring. † marks $d^* = 10$.

d	Acc	Variance	Dip	Silhouette	BC	Disagree
1	1.0000	0.1252	0.0567	0.2123	0.6885	0.5200
2	1.0000	0.1378	0.0477	0.2077	0.5584	0.7200
3	0.9400	0.1420	0.0477	0.1551	0.4720	0.7000
4	1.0000	0.1385	0.0726	0.1631	0.5721	0.8600
5	0.9400	0.1377	0.0692	0.1927	0.6100	0.8400
6	0.7800	0.1464	0.0463	0.1492	0.5482	0.7600
7	0.8800	0.1323	0.0778	0.1678	0.5233	0.8200
8	0.7200	0.1541	0.0425	0.2442	0.5141	0.8200
9	1.0000	0.1403	0.0594	0.1613	0.6427	0.8800
10	0.7400	0.1462	0.0506	0.1572	0.6277	0.9000
11	0.5800	0.1487	0.0481	0.1411	0.4741	0.8200
12	0.8600	0.1526	0.0670	0.1652	0.5822	0.8800
13	0.5400	0.1989	0.0497	0.1541	0.4619	0.8400
14 †	0.4000	0.1544	0.0648	0.1642	0.5744	0.8600
15	0.8200	0.1668	0.0758	0.1779	0.5632	0.9400
16	0.5800	0.1603	0.0373	0.1379	0.4559	0.8600
17	0.5400	0.1788	0.0610	0.1247	0.4937	0.9200
18	0.3600	0.1792	0.0447	0.1752	0.5203	0.9400
19	0.4800	0.1712	0.0345	0.1569	0.4998	0.8800
20	0.2600	0.1992	0.0359	0.1304	0.4410	0.9600

Table 5: CSD indicators for gemini-2.0-flash-001 on graph coloring. † marks $d^* = 14$.

d	Acc	Variance	Dip	Silhouette	BC	Disagree
1	1.0000	0.0625	0.0612	0.2702	0.6896	0.5600
2	1.0000	0.0810	0.0483	0.2424	0.7325	0.7200
3	0.8600	0.0815	0.0383	0.1890	0.5503	0.6200
4	1.0000	0.0673	0.0371	0.1071	0.4775	0.7800
5	0.9000	0.0860	0.0314	0.0646	0.7917	0.6800
6	0.6400	0.0742	0.0437	0.1362	0.5696	0.8400
7	0.7400	0.0715	0.0368	0.3485	0.7058	0.8200
8	0.7800	0.0670	0.0327	0.2821	0.7681	0.8200
9	0.7400	0.0597	0.0641	0.1069	0.5375	0.8800
10	0.6000	0.0697	0.0374	0.2737	0.7924	0.8600
11 †	0.2400	0.0591	0.0388	0.1198	0.4445	0.8800
12	0.6200	0.0655	0.0304	0.4340	0.8071	0.8400
13	0.2400	0.0541	0.0412	0.1057	0.4599	0.8400
14	0.1400	0.0890	0.0460	0.2628	0.7008	0.9200
15	0.6400	0.0730	0.0440	0.1292	0.4967	0.9000
16	0.4600	0.0817	0.0407	0.1046	0.6334	0.9000
17	0.6000	0.0927	0.0334	0.1483	0.3068	0.9200
18	0.3000	0.0875	0.0365	0.3444	0.7710	0.8800
19	0.1600	0.0936	0.0389	0.3599	0.5600	0.9400
20	0.1800	0.1131	0.0640	0.3472	0.7876	0.9400

Table 6: CSD indicators for gemini-2.0-flash-lite-001 on graph coloring. † marks $d^* = 11$.

2 Appendix B: Feature Ablation Study

This appendix reports the complete ablation study examining the contribution of CSD features to boundary detection. Table S1 presents the full matrix of 5 ablation conditions $\times$ 3 classifiers $\times$ 3 cross-validation schemes = 45 cells, each with F1 score and 95% bootstrap confidence intervals.

Table 7: Table S1: Full ablation matrix. F1 [95% CI] across ablation variants, classifiers, and CV schemes.

Ablation	Classifier	CV	F1 [95% CI]
1_pure_csd	logreg	lomo	0.670 [0.554, 0.768]
	logreg	lopo	0.670 [0.561, 0.769]
	logreg	loto	0.567 [0.457, 0.672]
	rf	lomo	0.690 [0.588, 0.786]
	rf	lopo	0.653 [0.549, 0.752]
	rf	loto	0.592 [0.489, 0.684]
	svm	lomo	0.591 [0.471, 0.702]
	svm	lopo	0.594 [0.475, 0.698]
	svm	loto	0.354 [0.242, 0.466]
2_csd_dynamics	logreg	lomo	0.734 [0.633, 0.825]
	logreg	lopo	0.708 [0.608, 0.797]
	logreg	loto	0.627 [0.532, 0.727]
	rf	lomo	0.727 [0.630, 0.815]
	rf	lopo	0.684 [0.582, 0.779]
	rf	loto	0.621 [0.524, 0.710]
	svm	lomo	0.645 [0.537, 0.744]
	svm	lopo	0.654 [0.544, 0.750]
	svm	loto	0.596 [0.503, 0.687]
3_difficulty_only	logreg	lomo	0.864 [0.788, 0.927]
	logreg	lopo	0.863 [0.790, 0.926]
	logreg	loto	0.791 [0.702, 0.875]
	rf	lomo	0.824 [0.735, 0.903]
	rf	lopo	0.822 [0.743, 0.895]
	rf	loto	0.810 [0.729, 0.887]
	svm	lomo	0.886 [0.821, 0.946]
	svm	lopo	0.884 [0.817, 0.939]
	svm	loto	0.791 [0.702, 0.875]

Table 8: Table S2: Incremental feature contribution to boundary detection F1.

Direction	Feature Added	New F1	Marginal Gain
forward_from_difficulty	dip_statistic	0.8485	+0.0270
forward_from_difficulty	csd_variance	0.8355	+0.0140
forward_from_difficulty	bimodality_coefficient	0.8353	+0.0138
forward_from_difficulty	disagreement_rate	0.8350	+0.0135
forward_from_difficulty	silhouette_k2	0.8324	+0.0109
forward_from_csd	difficulty_level	0.8143	+0.1617
forward_from_csd	difficulty_level_squared	0.8070	+0.1544
forward_from_csd	relative_position_in_range	0.8538	+0.2011

Table 9: Table S3: Permutation test results for CSD feature informativeness.

Variant	Unpermuted F1	Permuted F1	Drop	p-value
full_model	0.9546	0.9671	-0.0125	1.0000
pure_csd	0.6526	0.4889	+0.1638	0.0100

3 Appendix C: Model Routing Simulation Extended Results

This appendix presents the full routing simulation results across 2 tasks $\times$ 4 policies $\times$ 5 batch sizes $\times$ 3 difficulty distributions = 120 cells. The CSD-monitored routing policy uses real-time CSD indicator monitoring to decide when to escalate from a cheap model to a capable model.

Headline: CSD routing achieves 53.7% of oracle improvement at 58.5% of capable cost

3.1 Arithmetic

3.2 Graph Coloring

Table 10: Table SC: Routing results for Arithmetic.

Policy	B	Dist	Accuracy	Cost	Oracle Gap	Error Red.
always_cheap	10	uniform	0.5948	1.00	0.0%	0.0%
always_capable	10	uniform	0.6855	10.00	79.5%	22.4%
oracle	10	uniform	0.7090	8.05	100.0%	28.2%
csd_monitoring	10	uniform	0.6270	4.75	28.2%	7.9%
always_cheap	15	uniform	0.5930	1.00	0.0%	0.0%
always_capable	15	uniform	0.6881	10.00	83.0%	23.4%
oracle	15	uniform	0.7077	8.05	100.0%	28.2%
csd_monitoring	15	uniform	0.6107	3.32	15.4%	4.3%
always_cheap	20	uniform	0.5947	1.00	0.0%	0.0%
always_capable	20	uniform	0.6873	10.00	79.7%	22.9%
oracle	20	uniform	0.7109	8.04	100.0%	28.7%
csd_monitoring	20	uniform	0.6167	3.60	19.0%	5.4%
always_cheap	30	uniform	0.5934	1.00	0.0%	0.0%
always_capable	30	uniform	0.6888	10.00	81.6%	23.5%
oracle	30	uniform	0.7103	8.05	100.0%	28.8%
csd_monitoring	30	uniform	0.6121	3.63	16.1%	4.6%
always_cheap	50	uniform	0.5931	1.00	0.0%	0.0%
always_capable	50	uniform	0.6870	10.00	81.0%	23.1%
oracle	50	uniform	0.7090	8.07	100.0%	28.5%
csd_monitoring	50	uniform	0.6026	3.04	8.2%	2.3%
always_cheap	10	beta_easy	0.5828	1.00	0.0%	0.0%
always_capable	10	beta_easy	0.7061	10.00	90.1%	29.6%
oracle	10	beta_easy	0.7196	9.36	100.0%	32.8%
csd_monitoring	10	beta_easy	0.6183	3.69	26.0%	8.5%
always_cheap	15	beta_easy	0.5828	1.00	0.0%	0.0%
always_capable	15	beta_easy	0.7082	10.00	92.6%	30.1%
oracle	15	beta_easy	0.7182	9.37	100.0%	32.5%
csd_monitoring	15	beta_easy	0.6200	4.05	27.5%	8.9%
always_cheap	20	beta_easy	0.5835	1.00	0.0%	0.0%
always_capable	20	beta_easy	0.7085	10.00	92.2%	30.0%
oracle	20	beta_easy	0.7191	9.38	100.0%	32.6%
csd_monitoring	20	beta_easy	0.6092	3.32	19.0%	6.2%
always_cheap	30	beta_easy	0.5814	1.00	0.0%	0.0%
always_capable	30	beta_easy	0.7085	10.00	92.1%	30.4%
oracle	30	beta_easy	0.7195	9.38	100.0%	33.0%
csd_monitoring	30	beta_easy	0.5971	2.91	11.4%	3.8%
always_cheap	50	beta_easy	0.5804	1.00	0.0%	0.0%
always_capable	50	beta_easy	0.7066	10.00	92.4%	30.1%
oracle	50	beta_easy	0.7169	9.37	100.0%	32.5%
csd_monitoring	50	beta_easy	0.5986	3.16	13.4%	4.3%
always_cheap	10	beta_hard	0.6046	1.00	0.0%	0.0%
always_capable	10	beta_hard	0.6550	10.00	54.4%	12.8%
oracle	10	beta_hard	0.6973	6.29	100.0%	23.4%
csd_monitoring	10	beta_hard	0.6340	6.35	31.7%	7.4%
always_cheap	15	beta_hard	0.6036	1.00	0.0%	0.0%
always_capable	15	beta_hard	0.6541	10.00	53.5%	12.7%

Table 11: Table SC: Routing results for Graph Coloring.

Policy	B	Dist	Accuracy	Cost	Oracle Gap	Error Red.
always_cheap	10	uniform	0.5919	1.00	0.0%	0.0%
always_capable	10	uniform	0.7239	10.00	93.8%	32.3%
oracle	10	uniform	0.7326	7.87	100.0%	34.5%
csd_monitoring	10	uniform	0.7217	8.06	92.3%	31.8%
always_cheap	15	uniform	0.5943	1.00	0.0%	0.0%
always_capable	15	uniform	0.7244	10.00	93.9%	32.1%
oracle	15	uniform	0.7329	7.86	100.0%	34.2%
csd_monitoring	15	uniform	0.7187	8.03	89.7%	30.7%
always_cheap	20	uniform	0.5907	1.00	0.0%	0.0%
always_capable	20	uniform	0.7271	10.00	94.9%	33.3%
oracle	20	uniform	0.7345	7.89	100.0%	35.1%
csd_monitoring	20	uniform	0.7179	8.10	88.5%	31.1%
always_cheap	30	uniform	0.5925	1.00	0.0%	0.0%
always_capable	30	uniform	0.7248	10.00	94.1%	32.5%
oracle	30	uniform	0.7330	7.87	100.0%	34.5%
csd_monitoring	30	uniform	0.7195	8.07	90.4%	31.2%
always_cheap	50	uniform	0.5909	1.00	0.0%	0.0%
always_capable	50	uniform	0.7237	10.00	95.0%	32.5%
oracle	50	uniform	0.7307	7.85	100.0%	34.2%
csd_monitoring	50	uniform	0.7175	8.19	90.5%	30.9%
always_cheap	10	beta_easy	0.7678	1.00	0.0%	0.0%
always_capable	10	beta_easy	0.8661	10.00	94.1%	42.4%
oracle	10	beta_easy	0.8722	7.35	100.0%	45.0%
csd_monitoring	10	beta_easy	0.8553	6.39	83.8%	37.7%
always_cheap	15	beta_easy	0.7666	1.00	0.0%	0.0%
always_capable	15	beta_easy	0.8669	10.00	93.7%	42.9%
oracle	15	beta_easy	0.8736	7.35	100.0%	45.8%
csd_monitoring	15	beta_easy	0.8513	6.48	79.2%	36.3%
always_cheap	20	beta_easy	0.7687	1.00	0.0%	0.0%
always_capable	20	beta_easy	0.8661	10.00	94.0%	42.1%
oracle	20	beta_easy	0.8723	7.34	100.0%	44.8%
csd_monitoring	20	beta_easy	0.8519	6.44	80.3%	36.0%
always_cheap	30	beta_easy	0.7670	1.00	0.0%	0.0%
always_capable	30	beta_easy	0.8667	10.00	93.8%	42.8%
oracle	30	beta_easy	0.8732	7.36	100.0%	45.6%
csd_monitoring	30	beta_easy	0.8540	6.32	81.9%	37.3%
always_cheap	50	beta_easy	0.7663	1.00	0.0%	0.0%
always_capable	50	beta_easy	0.8672	10.00	96.5%	43.2%
oracle	50	beta_easy	0.8709	7.38	100.0%	44.8%
csd_monitoring	50	beta_easy	0.8524	6.44	82.3%	36.8%
always_cheap	10	beta_hard	0.4300	1.00	0.0%	0.0%
always_capable	10	beta_hard	0.5994	10.00	97.6%	29.7%
oracle	10	beta_hard	0.6036	8.72	100.0%	30.4%
csd_monitoring	10	beta_hard	0.5980	10.03	96.8%	29.5%
always_cheap	15	beta_hard	0.4313	1.00	0.0%	0.0%
always_capable	15	beta_hard	0.5965	10.00	94.8%	29.1%

4 Appendix D: Cross-Task Transfer Analysis

This appendix quantifies the distributional shift of CSD features across task families (arithmetic vs. graph coloring), measuring whether CSD indicators transfer across tasks using Kolmogorov-Smirnov tests and Cohen's d effect sizes.

Table 12: Table S4: CSD feature distributional shift across tasks (arithmetic vs. graph coloring).

Feature	Wasserstein	KS stat	KS p	Cohen's d	Overlap
csd_variance	0.0843	0.838	0.0000	2.286	0.180
dip_statistic	0.0369	0.750	0.0000	1.994	0.238
silhouette_k2	0.0957	0.762	0.0000	1.504	0.179
bimodality_coefficient	0.0413	0.188	0.2720	0.239	0.611
disagreement_rate	0.0693	0.458	0.0000	-0.065	0.171

Best new LOTO F1: 0.913194, Baseline LOTO F1 (zt_rf): 0.448

5 Appendix E: Syllogistic Logic Extended Analysis

This appendix presents the extended CSD analysis for syllogistic logic tasks across 3 weak LLMs and 22 difficulty levels ($d = 2$–30).

Table 13: Table S5: Syllogistic CSD model summaries.

Model	d^*	α	R^2	Flickering	Consensus
ministral-3b-2512	22	0.0460	0.0821	Yes	LT=20
llama-3.1-8b-instruct	N/A	N/A	N/A	Yes	LT=29
gemini-2.0-flash-lite-001	N/A	N/A	N/A	Yes	LT=29

5.1 ministral-3b-2512

Table 14: CSD indicators for ministral-3b-2512 on syllogistic logic.

d	Acc	Variance	Dip	Sil	BC	Disagree	Ashman D
2	1.0000	0.4573	0.0799	0.2568	0.7878	0.4000	10.2705
3	0.9800	0.4926	0.0759	0.2852	0.5641	0.4200	5.5765
4	0.9600	0.3853	0.0689	0.2629	0.6559	0.3600	6.5409
5	1.0000	0.3943	0.0676	0.2300	0.6026	0.4000	3.8124
6	0.9800	0.4014	0.1367	0.2238	0.5702	0.4200	5.4782
7	1.0000	0.3852	0.0812	0.2248	0.5830	0.4000	2.4682
8	1.0000	0.3595	0.0737	0.2180	0.5830	0.4000	4.4077
9	0.9600	0.3604	0.0949	0.2231	0.6229	0.4000	5.0993
10	0.9400	0.3620	0.1103	0.2109	0.5668	0.3400	4.6424
11	0.8800	0.3469	0.0748	0.2290	0.7041	0.4400	7.8330
12	0.7400	0.3575	0.1101	0.2048	0.5505	0.4200	4.8449
13	0.8800	0.3462	0.0942	0.1855	0.6097	0.4800	5.7308
14	0.8600	0.3345	0.0910	0.2077	0.6532	0.4490	5.9396
15	0.9000	0.3870	0.0706	0.2119	0.5577	0.4600	5.0554
16	0.8600	0.3875	0.1024	0.1842	0.6305	0.3400	5.2221
18	0.7400	0.3701	0.0614	0.2133	0.7132	0.3673	7.3278
20	0.7400	0.3973	0.0535	0.1759	0.4461	0.4694	4.7851
22	0.3800	0.3497	0.0520	0.1459	0.4895	0.4792	3.5531
24	0.6800	0.3738	0.0513	0.1831	0.4352	0.2800	4.4015
26	0.7200	0.4111	0.0650	0.1765	0.3866	0.4000	2.8002
28	0.6200	0.4105	0.1017	0.1678	0.5877	0.4490	5.3541
30	0.4600	0.4143	0.0632	0.1515	0.4884	0.2653	3.7957

5.2 llama-3.1-8b-instruct

5.3 gemini-2.0-flash-lite-001

Table 15: CSD indicators for llama-3.1-8b-instruct on syllogistic logic.

d	Acc	Variance	Dip	Sil	BC	Disagree	Ashman D
2	0.9200	0.5438	0.0841	0.2999	0.7258	0.3200	8.9806
3	0.7600	0.5395	0.0865	0.3153	0.7378	0.3600	8.7491
4	0.8600	0.4282	0.1156	0.2935	0.6231	0.3400	2.3523
5	0.8000	0.4465	0.1507	0.2423	0.7048	0.4400	6.6837
6	0.9400	0.4535	0.1305	0.2484	0.6200	0.3800	6.1564
7	0.9000	0.4539	0.0739	0.2148	0.5385	0.4082	6.1626
8	0.8600	0.4116	0.0727	0.2097	0.4423	0.3878	2.0035
9	0.8800	0.4327	0.0722	0.2224	0.5760	0.4375	5.6467
10	0.7000	0.4133	0.0832	0.1996	0.5449	0.4082	4.4896
11	0.8800	0.4004	0.0758	0.1762	0.4566	0.4000	3.2947
12	0.6400	0.4497	0.1010	0.1720	0.5664	0.4043	4.5988
13	0.8800	0.3886	0.0491	0.1668	0.5883	0.3600	2.5800
14	0.7600	0.3918	0.0795	0.1721	0.5923	0.3750	4.8857
15	0.8000	0.4207	0.0557	0.1499	0.6155	0.3125	4.8536
16	0.7200	0.3843	0.0636	0.1858	0.7996	0.4583	9.1229
18	0.7200	0.4098	0.0386	0.1787	0.7523	0.3333	5.2891
20	0.7400	0.4082	0.0818	0.1435	0.5833	0.4565	4.4098
22	0.7400	0.3510	0.0489	0.1049	0.4788	0.3778	3.0533
24	0.5200	0.4298	0.0615	0.1357	0.6318	0.4545	5.2663
26	0.7000	0.4924	0.0515	0.1393	0.4324	0.5000	2.4431
28	0.7200	0.4390	0.0738	0.1317	0.5410	0.4894	4.0775
30	0.5800	0.4087	0.0551	0.1160	0.4948	0.3864	3.4957

Table 16: CSD indicators for gemini-2.0-flash-lite-001 on syllogistic logic.

d	Acc	Variance	Dip	Sil	BC	Disagree	Ashman D
2	1.0000	0.4998	0.0923	0.3591	0.7626	0.4000	7.1515
3	1.0000	0.5099	0.0889	0.3640	0.7012	0.4000	7.3133
4	1.0000	0.3834	0.0730	0.3102	0.6058	0.4000	2.3928
5	1.0000	0.4277	0.1435	0.2769	0.5331	0.4000	3.9884
6	1.0000	0.3915	0.1543	0.2723	0.5868	0.4000	5.5866
7	1.0000	0.4090	0.0678	0.2552	0.5122	0.4000	5.4535
8	1.0000	0.4366	0.0712	0.2155	0.4841	0.4000	3.3312
9	1.0000	0.4172	0.0786	0.2316	0.6154	0.4000	2.3987
10	1.0000	0.4236	0.0594	0.1810	0.6906	0.4000	6.6654
11	1.0000	0.4176	0.0727	0.1920	0.5412	0.4000	3.4424
12	0.9200	0.4147	0.0716	0.2178	0.6114	0.4694	5.7395
13	0.9800	0.4158	0.0412	0.1931	0.5345	0.3800	3.3051
14	0.9800	0.4086	0.0744	0.2194	0.5326	0.4200	4.7152
15	1.0000	0.4163	0.0444	0.1882	0.4993	0.4000	2.6434
16	0.9800	0.4258	0.0768	0.2418	0.8377	0.4200	11.1571
18	0.9400	0.4050	0.0505	0.1685	0.4808	0.4600	2.7477
20	0.9800	0.4114	0.0680	0.1663	0.6156	0.4082	4.3927
22	0.9400	0.3900	0.0597	0.1514	0.5404	0.4600	3.7915
24	0.9800	0.3800	0.0860	0.2249	0.6887	0.4200	5.6603
26	1.0000	0.4202	0.0694	0.1953	0.6457	0.4000	5.2032
28	0.9800	0.3956	0.0668	0.1839	0.6606	0.4200	6.7514
30	0.9800	0.3638	0.0565	0.1810	0.5301	0.4200	3.6729

6 Appendix F: Sample-Size Sensitivity Analysis

This appendix examines how boundary detection F1 degrades as the number of sampled responses per difficulty level (N) decreases from 50 to 10.

Table 17: Table S6: Boundary detection F1 and dip detection rate as function of sample size N.

N	F1	Dip Rate
10	0.8120	0.1402
15	0.8656	0.1584
20	0.8848	0.1726
25	0.9006	0.1846
30	0.9175	0.1950
40	0.9411	0.2126
50	0.9419	0.2273

Minimum viable N for F1>0.80: 15.0

7 Appendix G: Negative Control Analysis

This appendix reports the false positive rate of CSD indicators in regions where the model is safely within its capability range (far from d^*). Indicators should *not* show significant trends in safe regions.

Table 18: Table S7: Negative control and consistency metrics.

Metric	Value
Negative control pass rate	0.1333
Consistency cells tested	30
Fraction significant	0.2667
Fraction correct direction	0.1333

Table 19: Table S8: Effect sizes for CSD indicators between safe and near-boundary regions.

Indicator	Cohen's d	Cliff's δ
variance	-0.2115	-0.0856
dip_statistic	-0.5292	-0.2130
silhouette_k2	-0.7081	-0.3245
bimodality_coefficient	-0.5379	-0.2026
disagreement_rate	0.8803	0.4570

8 Appendix H: Sampling Temperature Configuration

This appendix documents the sampling temperature used for all CSD experiments and provides a breakdown of response generation parameters across tasks.

8.1 Arithmetic

Table 20: Table S9: Sampling parameters for Arithmetic.

Parameter	Value
Temperature	0.8
Top-p	0.95
Max tokens	2048
Difficulty levels	[2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25]
Responses/level	50
Models	llama-3.1-8b-instruct, gemini-2.0-flash-001, gpt-4o-mini

8.2 Graph Coloring

Table 21: Table S9: Sampling parameters for Graph Coloring.

Parameter	Value
Temperature	0.8
Top-p	N/A
Max tokens	N/A
Difficulty levels	20
Responses/level	N/A
Models	gpt-4o-mini, gemini-2.0-flash-001, gemini-2.0-flash-lite-001

9 Supplementary Figures

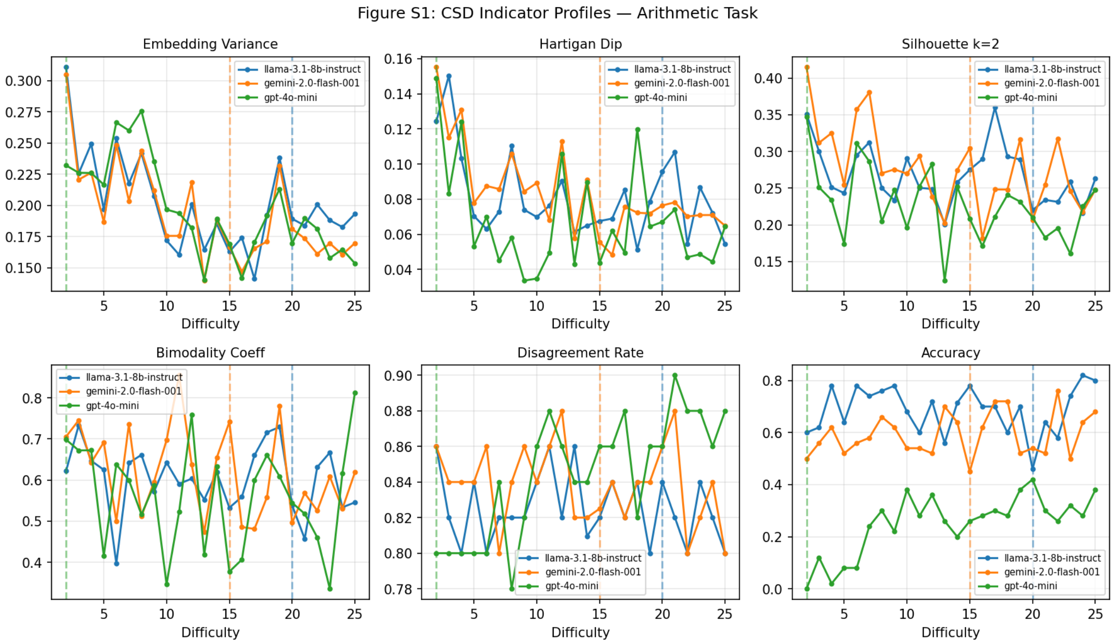

Figure 1: CSD Indicator Profiles – Arithmetic Task

Figure 2: CSD Indicator Profiles – Graph Coloring Task

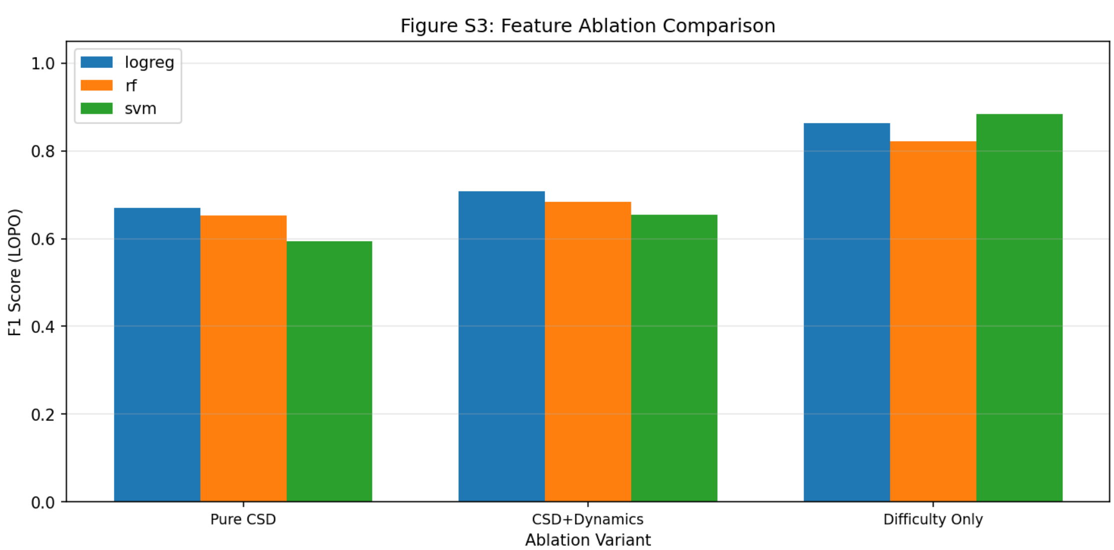

Figure 3: Feature Ablation Comparison

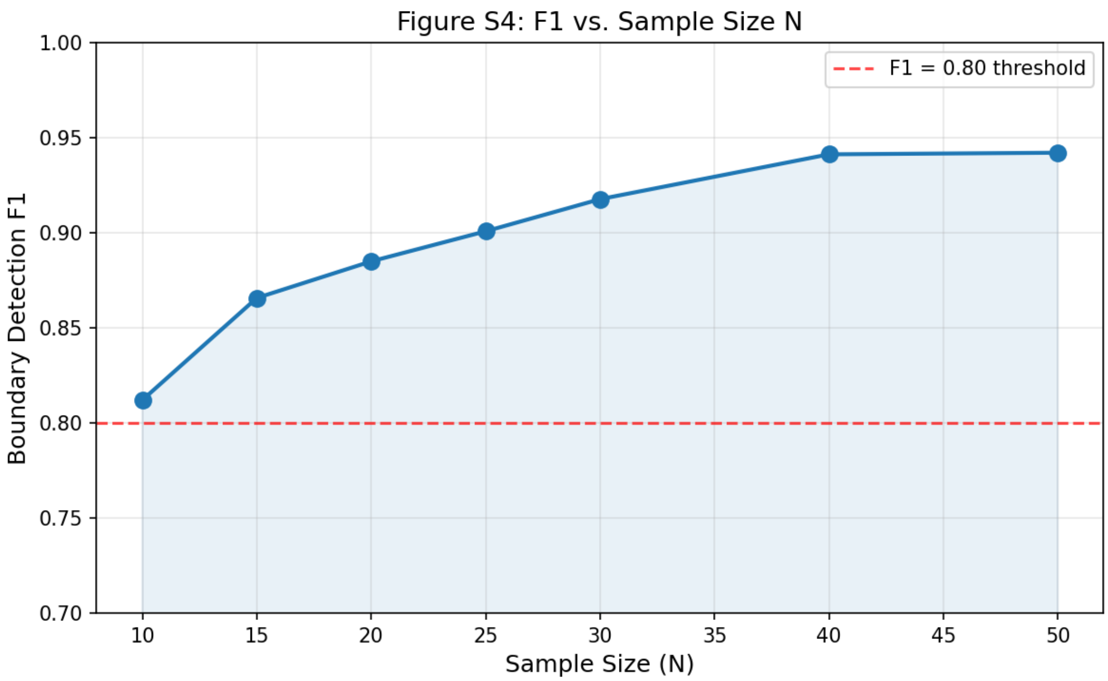

Figure 4: F1 vs. Sample Size N